

ANALYSIS OF AUTHENTICATION AND USER SESSION ACTIVITY IN KALI LINUX

Capstone Project Report

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Domain: Cyber Security

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DECLARATION

I hereby declare that the capstone project report entitled 'ANALYSIS OF AUTHENTICATION AND USER SESSION ACTIVITY IN KALI LINUX' submitted to Launched Global and Gujarat Vidyapith, Ahmedabad, is an authentic record of independent research and practical work carried out by me under professional supervision.

The data, log analyses, session records, and security interpretations presented in this report are genuine and derived directly from the forensic execution of security auditing tools within the designated environment. This work has not been submitted previously to any other university or institute for the award of any degree or diploma.

ACKNOWLEDGEMENT

I express my sincere gratitude to Launched Global for offering me the opportunity to undergo hands-on practical training in Cyber Security. This capstone project provided an invaluable platform to develop forensic skills in Linux system auditing, authentication mechanism analysis, and log file database structure inspection.

I extend my heartfelt thanks to my academic mentors at Gujarat Vidyapith, Ahmedabad, for their constant encouragement, academic foundation, and guidance throughout the second semester of the MSc IT program.

ABSTRACT

Log file analysis forms the baseline of digital forensics and security operations in Unix-like operating systems. Modern Linux distributions, such as Kali Linux, have transitioned from traditional flat-binary log files toward database-driven and journaled logging mechanisms. This project delivers a thorough investigation into authentication event logs and user session records within a Kali Linux rolling environment.

1. INTRODUCTION

Linux operating systems record system events, service activities, and user authorization sequences in centralized logging repositories. In cyber security and digital forensics, log analysis is indispensable for detecting unauthorized access, investigating security breaches, and maintaining compliance standards.

2. PROBLEM STATEMENT

As Linux environments transition from traditional binary files (`/var/log/wtmp`) to modern database-driven frameworks (`wtmpdb` using SQLite) and centralized journal services (`systemd-journald`), standard forensic practices must be adapted.

3. PROJECT OBJECTIVES

The principal objectives of this capstone project are log file identification, structural file analysis, database schema extraction, binary and utility mapping, session timeline analysis, active session monitoring, authentication event parsing, and security interpretation.

4. TOOLS AND TECHNOLOGIES

The analysis was executed using standard Linux administrative, auditing, and database inspection tools including Kali Linux, VirtualBox, journalctl, grep, wtmpdb, last, sqlite3, who, w, and users.

5. SYSTEM ENVIRONMENT

The practical investigation was conducted on a virtualized Kali Linux operating system. Initial workspace preparation, evidence directory creation, and system kernel startup logs:

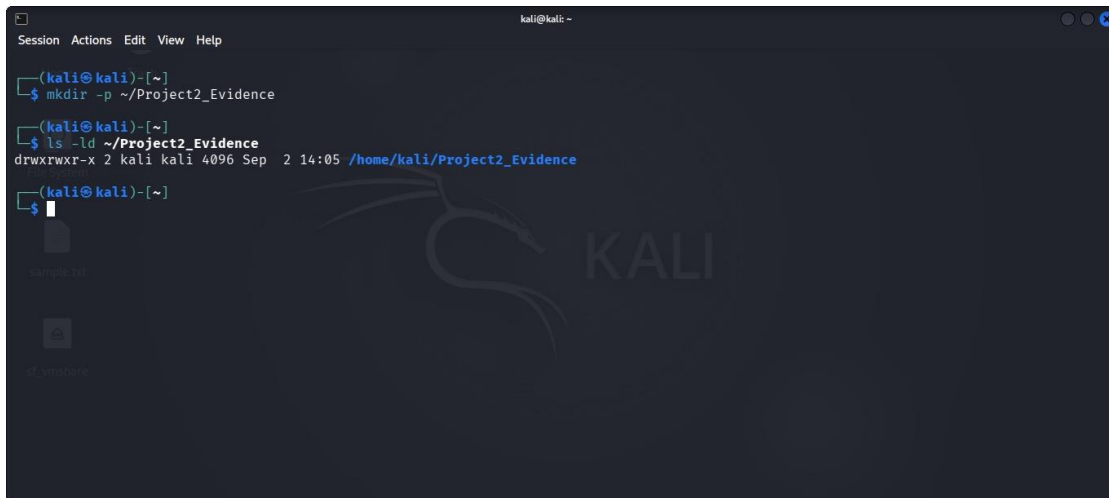
A screenshot of a Kali Linux terminal window. The window has a title bar with 'kali@kali: ~' and standard window controls. The terminal shows a series of commands and their outputs. First, the user runs 'mkdir -p ~/Project2_Evidence'. Then, they run 'ls -ld ~/Project2_Evidence', which outputs 'drwxrwxr-x 2 kali kali 4096 Sep 2 14:05 /home/kali/Project2_Evidence'. The terminal background features a large, faint 'KALI' logo. The prompt is '(kali@kali)-[~]'.

Figure 1: Creation of the ~/Project2_Evidence directory for artifact collection.

```
kali@kali: ~  
Session Actions Edit View Help  
--(kali@kali)-[~]  
$ date  
Wed Sep 2 02:02:09 PM EDT 2026  
  
--(kali@kali)-[~]  
$ timedatectl  
Local time: Wed 2026-09-02 14:02:17 EDT  
Universal time: Wed 2026-09-02 18:02:17 UTC  
RTC time: Wed 2026-09-02 18:02:17  
Time zone: America/New_York (EDT, -0400)  
System clock synchronized: no  
NTP service: inactive  
RTC in local TZ: no  
  
--(kali@kali)-[~]  
$
```

Figure 2: Kernel version inspection and VirtualBox hypervisor DMI identification.

```
kali@kali: ~  
Session Actions Edit View Help  
--(kali@kali)-[~]  
$ sudo wtmpdb last --ip  
kali tty7 :0 Wed Sep 2 12:21 - still logged in  
lightdm tty7 :0 Wed Sep 2 12:21 - 12:21 (00:00)  
kali tty7 :0 Wed Sep 2 11:31 - still logged in  
lightdm tty7 :0 Wed Sep 2 11:31 - 11:31 (00:00)  
kali tty7 :0 Wed Sep 2 03:00 - still logged in  
lightdm tty7 :0 Wed Sep 2 02:50 - 03:00 (00:09)  
kali tty7 :0 Tue Sep 1 23:24 - still logged in  
lightdm tty7 :0 Tue Sep 1 23:23 - 23:24 (00:01)  
kali pts/4 127.0.0.1 Tue Sep 1 13:44 - still logged in  
kali pts/3 ::1 Tue Sep 1 13:25 - still logged in  
kali tty7 :0 Tue Sep 1 12:51 - still logged in  
lightdm tty7 :0 Tue Sep 1 12:51 - 12:51 (00:00)  
kali tty7 :0 Tue Sep 1 12:49 - still logged in  
lightdm tty7 :0 Tue Sep 1 12:49 - 12:49 (00:00)  
kali tty7 :0 Tue Sep 1 12:40 - still logged in  
lightdm tty7 :0 Tue Sep 1 12:40 - 12:40 (00:00)  
kali tty7 :0 Tue Sep 1 06:16 - still logged in  
lightdm tty7 :0 Tue Sep 1 06:15 - 06:16 (00:00)  
kali tty7 :0 Thu Aug 27 11:26 - still logged in  
lightdm tty7 :0 Thu Aug 27 11:26 - 11:26 (00:00)  
kali tty7 :0 Thu Aug 27 11:21 - still logged in  
lightdm tty7 :0 Thu Aug 27 11:21 - 11:21 (00:00)  
kali tty7 :0 Tue Aug 18 11:41 - still logged in  
lightdm tty7 :0 Tue Aug 18 11:21 - 11:41 (00:20)  
kali tty7 :0 Tue Aug 18 00:08 - still logged in  
lightdm tty7 :0 Tue Aug 18 00:07 - 00:08 (00:00)  
kali tty7 :0 Mon Aug 17 11:33 - still logged in  
lightdm tty7 :0 Mon Aug 17 11:33 - 11:33 (00:00)  
kali tty7 :0 Mon Aug 17 07:24 - still logged in  
lightdm tty7 :0 Mon Aug 17 07:21 - 07:24 (00:02)
```

Figure 3: System clock and timezone verification via date and timedatectl.

```
kali@kali: ~  
Session Actions Edit View Help  
kali tty7 Mon Jul 20 06:08:42 2026 - still logged in :0  
lightdm tty7 Mon Jul 20 06:08:36 2026 - Mon Jul 20 06:08:42 2026 (00:00) :0  
kali tty7 Mon Jul 20 06:04:35 2026 - still logged in :0  
lightdm tty7 Mon Jul 20 06:04:25 2026 - Mon Jul 20 06:04:35 2026 (00:00) :0  
root pts/2 Mon Jul 20 05:35:13 2026 - still logged in :0  
kali tty7 Mon Jul 20 04:54:55 2026 - still logged in :0  
lightdm tty7 Mon Jul 20 04:54:41 2026 - Mon Jul 20 04:54:55 2026 (00:00) :0  
root pts/1 Sun Jul 19 11:27:26 2026 - Sun Jul 19 11:30:40 2026 (00:03)  
root pts/1 Sun Jul 19 11:27:08 2026 - Sun Jul 19 11:27:19 2026 (00:00)  
kali tty7 Fri Jul 17 02:04:53 2026 - still logged in :0  
lightdm tty7 Fri Jul 17 02:04:45 2026 - Fri Jul 17 02:04:53 2026 (00:00) :0  
kali tty7 Tue Jul 7 06:17:48 2026 - still logged in :0  
lightdm tty7 Tue Jul 7 06:17:37 2026 - Tue Jul 7 06:17:48 2026 (00:00) :0  
postgres Tue Jun 16 10:29:31 2026 - Tue Jun 16 10:29:31 2026 (00:00)  
  
wtmddb begins Tue Jun 16 10:29:31 2026  
  
(kali@kali)~  
$ who  
kali seat0 2026-09-02 11:31 (:0)  
  
(kali@kali)~  
$ w  
11:46:48 up 15 min, 1 user, load average: 0.05, 0.15, 0.23  
USER TTY FROM LOGIN# IDLE JCPU PCPU WHAT  
kali - 11:31 0.00s 0.01s lightdm --session-child 13 24  
  
(kali@kali)~  
$ users  
kali  
  
(kali@kali)~  
$
```

Figure 4: Complete OS release specifications, machine ID, and boot ID parameters.

6. LOG FILE IDENTIFICATION

Authentication and session logs in Kali Linux reside predominantly under the `/var/log/` hierarchy, including `auth.log`, `wtmpt`, `wtmpt.db`, `lastlog`, and `btmpt`.

7. FILE TYPE ANALYSIS

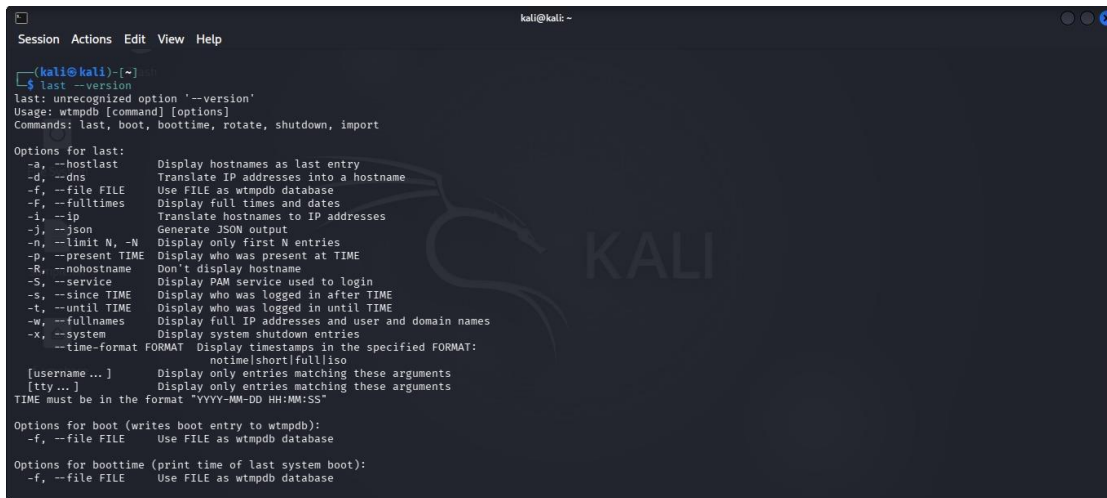
Verifying underlying file formats and directory structures for session tracking files:

```
kali@kali: ~  
Session Actions Edit View Help  
(kali@kali)~  
$ sudo journalctl -b --no-pager | head -50  
[sudo] password for kali:  
Sep 02 13:57:43 kali kernel: Linux version 6.19.14+kali-amd64 (devel@kali.org) (x86_64-linux-gnu-gcc-15 (Debian 15.2.0-17) 15.2.0, GNU ld (GNU Binutils for Debian) 2.46) #1 SMP PREEMPT_DYNAMIC Kali 6.19.14-1-kali1 (2026-05-05)  
Sep 02 13:57:43 kali kernel: Command line: BOOT_IMAGE=/boot/vmlinuz-6.19.14+kali-amd64 root=UUID=d8f71766-9813-406b-90f2-fd6cd673b79e ro quiet splash  
Sep 02 13:57:43 kali kernel: BIOS-provided physical RAM map:  
Sep 02 13:57:43 kali kernel: BIOS-e820: [mem 0x0000000000000000-0x000000000009fbff] usable  
Sep 02 13:57:43 kali kernel: BIOS-e820: [mem 0x000000000009fc00-0x00000000000fffff] reserved  
Sep 02 13:57:43 kali kernel: BIOS-e820: [mem 0x00000000000fa000-0x00000000000fffff] reserved  
Sep 02 13:57:43 kali kernel: BIOS-e820: [mem 0x0000000000100000-0x0000000007fffff] usable  
Sep 02 13:57:43 kali kernel: BIOS-e820: [mem 0x0000000007ff0000-0x0000000007fffff] ACPI data  
Sep 02 13:57:43 kali kernel: BIOS-e820: [mem 0x00000000fec00000-0x00000000fec0ffff] reserved  
Sep 02 13:57:43 kali kernel: BIOS-e820: [mem 0x00000000fee00000-0x00000000fee0ffff] reserved  
Sep 02 13:57:43 kali kernel: BIOS-e820: [mem 0x00000000fffc0000-0x00000000fffcffff] reserved  
Sep 02 13:57:43 kali kernel: NX (Execute Disable) protection: active  
Sep 02 13:57:43 kali kernel: APIC: Static calls initialized  
Sep 02 13:57:43 kali kernel: SMBIOS 2.5 present.  
Sep 02 13:57:43 kali kernel: DMI: innotek GmbH VirtualBox/VirtualBox, BIOS VirtualBox 12/01/2006  
Sep 02 13:57:43 kali kernel: DMI: Memory slots populated: 0/0  
Sep 02 13:57:43 kali kernel: Hypervisor detected: KVM  
Sep 02 13:57:43 kali kernel: last_pfn = 0x80000 max_arch_pfn = 0x400000000  
Sep 02 13:57:43 kali kernel: kvm-clock: Using msrc 4b564d01 and 4b564d00  
Sep 02 13:57:43 kali kernel: kvm-clock: using sched offset of 29239248865 cycles  
Sep 02 13:57:43 kali kernel: clocksource: kvm-clock: mask: 0xffffffffffffffff max_cycles: 0x1cd42e4dffb, max_idle_ns: 881590591483 ns  
Sep 02 13:57:43 kali kernel: tsc: Detected 2712.000 MHz processor  
Sep 02 13:57:43 kali kernel: e820: update [mem 0x00000000-0x000000ffff] usable ==> reserved  
Sep 02 13:57:43 kali kernel: e820: remove [mem 0x00000000-0x000000ffff] usable  
Sep 02 13:57:43 kali kernel: last_pfn = 0x80000 max_arch_pfn = 0x400000000  
Sep 02 13:57:43 kali kernel: MTRR map: 3 entries (3 fixed + 0 variable; max 19), built from 8 variable MTRRs  
Sep 02 13:57:43 kali kernel: x86/PAT: Configuration [0-7]: WB UC UC WB WP UC- WT  
Sep 02 13:57:43 kali kernel: CPU MTRRs: all blank - virtualized system.  
Sep 02 13:57:43 kali kernel: found SMP MP-table at [mem 0x0009fbf0-0x0009fbff]
```

Figure 5: Detailed file size and permission breakdown of `/var/log/` session tracking files.

8. WTMPDB AND LAST COMMAND ANALYSIS

Inspecting helper command options and utility bindings for wtmpdb and last:

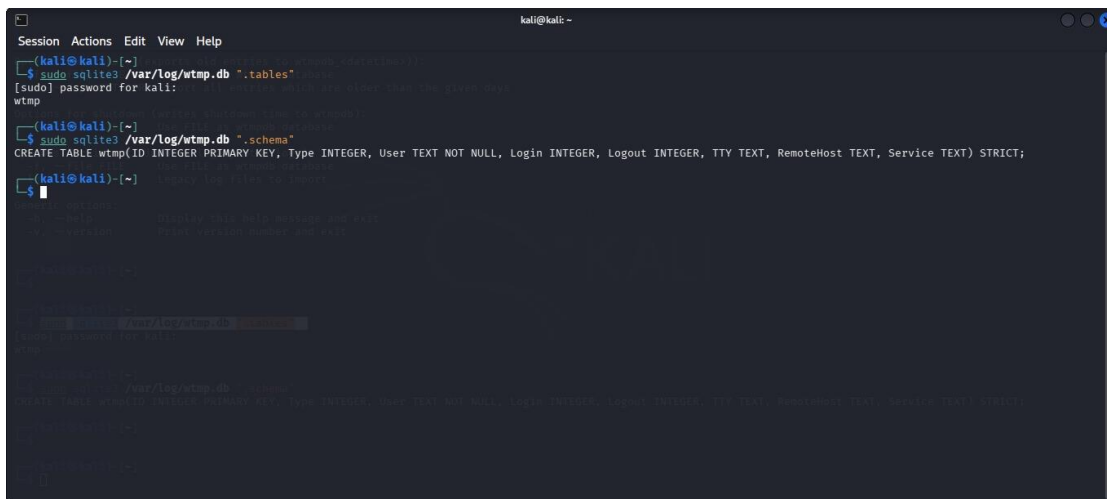


```
kali@kali: ~  
Session Actions Edit View Help  
[kali@kali]~  
$ last --version  
last: unrecognized option '--version'  
Usage: wtmpdb [command] [options]  
Commands: last, boot, boottime, rotate, shutdown, import  
  
Options for last:  
-a, --hostlast      Display hostnames as last entry  
-d, --dns           Translate IP addresses into a hostname  
-f, --file FILE     Use FILE as wtmpdb database  
-F, --fulltimes     Display full times and dates  
-i, --ip           Translate hostnames to IP addresses  
-j, --json          Generate JSON output  
-n, --limit N, -N   Display only first N entries  
-p, --present TIME  Display who was present at TIME  
-R, --nohostname    Don't display hostname  
-s, --service       Display PAM service used to login  
-S, --since TIME    Display who was logged in after TIME  
-t, --until TIME    Display who was logged in until TIME  
-w, --fullnames     Display full IP addresses and user and domain names  
-x, --system        Display system shutdown entries  
--time-format FORMAT Display timestamps in the specified FORMAT:  
                    notime|short|full|iso  
[username...]       Display only entries matching these arguments  
[tty...]            Display only entries matching these arguments  
TIME must be in the format "YYYY-MM-DD HH:MM:SS"  
  
Options for boot (writes boot entry to wtmpdb):  
-f, --file FILE     Use FILE as wtmpdb database  
  
Options for boottime (print time of last system boot):  
-f, --file FILE     Use FILE as wtmpdb database
```

Figure 6: Command-line reference documentation for wtmpdb subcommands and flags.

9. SQLITE DATABASE ANALYSIS

Interrogating the database schema directly via sqlite3 CLI:



```
kali@kali: ~  
Session Actions Edit View Help  
[kali@kali]~  
$ sudo sqlite3 /var/log/wtmp.db ".tables"  
[sudo] password for kali:  
wtmp  
[kali@kali]~  
$ sudo sqlite3 /var/log/wtmp.db ".schema"  
CREATE TABLE wtmp(ID INTEGER PRIMARY KEY, Type INTEGER, User TEXT NOT NULL, Login INTEGER, Logout INTEGER, TTY TEXT, RemoteHost TEXT, Service TEXT) STRICT;  
[kali@kali]~  
$
```

Figure 7: Interrogating /var/log/wtmp.db via sqlite3 to output table definitions.

10. HISTORICAL SESSION ANALYSIS

Chronological logs of user sessions and boot records extracted via wtmpdb last:

```
kali@kali: ~  
Session Actions Edit View Help  
[kali@kali]~$ sudo journalctl --no-pager | grep -E "session opened|session closed" > ~/Project2_Evidence/session_evidence.txt  
[kali@kali]~$ cat ~/Project2_Evidence/session_evidence.txt  
Jul 07 06:17:36 kali lightdm[883]: pam_unix(lightdm-greeter:session): session opened for user lightdm(uid=106) by (uid=0)  
Jul 07 06:17:37 kali (systemd)[902]: pam_unix(systemd-user:session): session opened for user lightdm(uid=106) by lightdm(uid=0)  
Jul 07 06:17:37 kali lightdm[883]: pam_unix(lightdm-greeter:session): session opened for user lightdm(uid=106) by (uid=0)  
Jul 07 06:17:48 kali lightdm[883]: pam_unix(lightdm-greeter:session): session closed for user lightdm  
Jul 07 06:17:48 kali lightdm[883]: pam_unix(lightdm-greeter:session): session closed for user lightdm  
Jul 07 06:17:48 kali lightdm[1062]: pam_unix(lightdm:session): session opened for user kali(uid=1000) by kali(uid=0)  
Jul 07 06:17:48 kali (systemd)[1086]: pam_unix(systemd-user:session): session opened for user kali(uid=1000) by kali(uid=0)  
Jul 07 02:04:44 kali lightdm[717]: pam_unix(lightdm-greeter:session): session opened for user lightdm(uid=106) by (uid=0)  
Jul 07 02:04:44 kali (systemd)[725]: pam_unix(systemd-user:session): session opened for user lightdm(uid=106) by lightdm(uid=0)  
Jul 07 02:04:45 kali lightdm[717]: pam_unix(lightdm-greeter:session): session opened for user lightdm(uid=106) by (uid=0)  
Jul 07 02:04:53 kali lightdm[717]: pam_unix(lightdm-greeter:session): session closed for user lightdm  
Jul 07 02:04:53 kali lightdm[717]: pam_unix(lightdm-greeter:session): session closed for user lightdm  
Jul 07 02:04:53 kali lightdm[883]: pam_unix(lightdm:session): session opened for user kali(uid=1000) by kali(uid=0)  
Jul 07 02:04:53 kali (systemd)[907]: pam_unix(systemd-user:session): session opened for user kali(uid=1000) by kali(uid=0)  
Jul 07 02:05:01 kali CRON[1547]: pam_unix(cron:session): session opened for user root(uid=0) by root(uid=0)  
Jul 07 02:05:01 kali CRON[1547]: pam_unix(cron:session): session closed for user root  
Jul 07 02:05:04 kali (sd-pam)[727]: pam_unix(systemd-user:session): session closed for user lightdm  
Jul 07 02:09:01 kali CRON[4054]: pam_unix(cron:session): session opened for user root(uid=0) by root(uid=0)  
Jul 07 02:09:01 kali CRON[4054]: pam_unix(cron:session): session closed for user root  
Jul 07 02:15:01 kali CRON[7383]: pam_unix(cron:session): session opened for user root(uid=0) by root(uid=0)  
Jul 07 02:15:01 kali CRON[7383]: pam_unix(cron:session): session closed for user root  
Jul 07 02:17:01 kali CRON[8336]: pam_unix(cron:session): session opened for user root(uid=0) by root(uid=0)  
Jul 07 02:17:01 kali CRON[8336]: pam_unix(cron:session): session closed for user root  
Jul 07 07:55:01 kali CRON[12703]: pam_unix(cron:session): session opened for user root(uid=0) by root(uid=0)  
Jul 07 07:55:01 kali CRON[12703]: pam_unix(cron:session): session closed for user root  
Jul 07 08:05:01 kali CRON[17829]: pam_unix(cron:session): session opened for user root(uid=0) by root(uid=0)  
Jul 07 08:05:01 kali CRON[17829]: pam_unix(cron:session): session closed for user root
```

Figure 8: Historical login tracking via wtmpdb across multiple system sessions.

```
kali@kali: ~  
Session Actions Edit View Help  
[kali@kali]~$ who  
kali    seat0      2026-09-02 12:21 (:0)  
[kali@kali]~$ w  
12:50:47 up 29 min,  1 user,  load average: 0.18, 0.27, 0.24  
USER      TTY      FROM          LOGIN@   IDLE   JCPU   PCPU   WHAT  
kali      -        -            12:21    -      0.00s  0.01s  lightdm --session-child 13 24  
[kali@kali]~$ users  
kali  
[kali@kali]~$
```

Figure 9: Additional session outputs detailing historical graphical entries.

```
kali@kali: ~  
Session Actions Edit View Help  
kali tty7 :0 Tue Jul 21 07:27:08 2026 - still logged in  
lightdm tty7 :0 Tue Jul 21 07:26:59 2026 - Tue Jul 21 07:27:08 2026 (00:00)  
kali :0 Mon Jul 20 11:42:56 2026 - still logged in  
kali tty7 :0 Mon Jul 20 11:26:16 2026 - still logged in  
lightdm tty7 :0 Mon Jul 20 11:25:46 2026 - Mon Jul 20 11:26:16 2026 (00:00)  
kali tty7 :0 Mon Jul 20 07:56:38 2026 - still logged in  
lightdm tty7 :0 Mon Jul 20 07:56:26 2026 - Mon Jul 20 07:56:37 2026 (00:00)  
kali tty7 :0 Mon Jul 20 07:32:07 2026 - still logged in  
lightdm tty7 :0 Mon Jul 20 07:31:20 2026 - Mon Jul 20 07:32:07 2026 (00:00)  
kali tty7 :0 Mon Jul 20 07:03:10 2026 - still logged in  
lightdm tty7 :0 Mon Jul 20 07:02:57 2026 - Mon Jul 20 07:03:10 2026 (00:00)  
kali tty7 :0 Mon Jul 20 06:58:20 2026 - still logged in  
lightdm tty7 :0 Mon Jul 20 06:58:08 2026 - Mon Jul 20 06:58:20 2026 (00:00)  
kali tty7 :0 Mon Jul 20 06:54:15 2026 - still logged in  
lightdm tty7 :0 Mon Jul 20 06:54:06 2026 - Mon Jul 20 06:54:15 2026 (00:00)  
kali tty7 :0 Mon Jul 20 06:21:22 2026 - still logged in  
lightdm tty7 :0 Mon Jul 20 06:21:12 2026 - Mon Jul 20 06:21:22 2026 (00:00)  
kali tty7 :0 Mon Jul 20 06:08:42 2026 - still logged in  
lightdm tty7 :0 Mon Jul 20 06:08:36 2026 - Mon Jul 20 06:08:42 2026 (00:00)  
kali tty7 :0 Mon Jul 20 06:04:35 2026 - still logged in  
lightdm tty7 :0 Mon Jul 20 06:04:25 2026 - Mon Jul 20 06:04:35 2026 (00:00)  
root pts/2 Mon Jul 20 05:35:13 2026 - still logged in  
kali tty7 :0 Mon Jul 20 04:54:55 2026 - still logged in  
lightdm tty7 :0 Mon Jul 20 04:54:44 2026 - Mon Jul 20 04:54:55 2026 (00:00)  
root pts/1 Sun Jul 19 11:27:26 2026 - Sun Jul 19 11:30:40 2026 (00:03)  
root pts/1 Sun Jul 19 11:27:08 2026 - Sun Jul 19 11:27:19 2026 (00:00)  
kali tty7 :0 Fri Jul 17 02:04:53 2026 - still logged in  
lightdm tty7 :0 Fri Jul 17 02:04:45 2026 - Fri Jul 17 02:04:53 2026 (00:00)  
kali tty7 :0 Tue Jul 7 06:17:48 2026 - still logged in  
lightdm tty7 :0 Tue Jul 7 06:17:37 2026 - Tue Jul 7 06:17:48 2026 (00:00)  
postgres Tue Jun 16 10:29:31 2026 - Tue Jun 16 10:29:31 2026 (00:00)  
wtmpdb begins Tue Jun 16 10:29:31 2026
```

Figure 10: Complete chronological list of user logins extracted from the database.

11. FULL TIMESTAMP ANALYSIS

High-resolution precision timestamp logs using wtmpdb last --fulltimes:

```
kali@kali: ~  
Session Actions Edit View Help  
drwxr-xr-x 2 _gvm _gvm 4.0K Apr 16 01:42 notus-scanner  
drwxr-xr-x 2 root root 4.0K Apr 27 17:25 openvpn  
drwxr-xr-x 2 root root 4.0K Jun 16 10:23 postgresql  
drwxr-xr-x 2 redis adm 4.0K Jun 16 10:29 redis  
drwxr-xr-x 4 root root 4.0K Jun 16 10:25 runit  
drwxr-xr-x 3 root adm 4.0K Jul 28 08:25 samba  
drwxr-xr-x 2 speech-dispatcher root 4.0K Jun 16 10:25 speech-dispatcher  
drwxr-xr-x 2 stunnel stunnel 4.0K Jun 16 10:28 stunnel  
drwxr-xr-x 2 root root 4.0K Jan 15 2024 sysstat  
-rw-r--r-- 1 root utmp 37K Sep 2 12:21 wtmp  
-rw-r--r-- 1 root root 16K Sep 2 12:21 wtmp.db  
-rw-r--r-- 1 root root 25K Sep 2 12:21 Xorg.0.Log  
-rw-r--r-- 1 root root 28K Sep 2 12:13 Xorg.0.Log.older  
  
kali@kali:~  
$ ls -lh /var/log/wtmp /var/log/wtmp.db /var/log/lastlog /var/log/btmp  
-rw-rw-r-- 1 root utmp 0 Aug 10 05:42 /var/log/btmp  
-rw-rw-r-- 1 root utmp 286K Sep 1 13:44 /var/log/lastlog  
-rw-rw-r-- 1 root utmp 37K Sep 2 12:21 /var/log/wtmp  
-rw-rw-r-- 1 root root 16K Sep 2 12:21 /var/log/wtmp.db  
  
kali@kali:~  
$ file /var/log/wtmp  
/var/log/wtmp: data  
  
kali@kali:~  
$ file /var/log/wtmp.db  
/var/log/wtmp.db: SQLite 3.x database, last written using SQLite version 3046001, file counter 155, database pages 4, cookie 0x1, schema 4, UTF-8, version-valid-for 15  
5  
  
kali@kali:~  
$
```

Figure 11: High-precision timestamp audit using sudo wtmpdb last --fulltimes.

12. CURRENT USER SESSION ANALYSIS

Real-time inspection of active user sessions using who, w, and users utilities:


```
kali@kali: ~  
Session Actions Edit View Help  
Sep 02 12:52:35 kali sudo(25156): pam_unix(sudo:session): session closed for user root  
Sep 02 12:53:25 kali sudo(26057): pam_unix(sudo:session): session opened for user root(uid=0) by kali(uid=1000)  
Sep 02 12:53:27 kali sudo(26057): pam_unix(sudo:session): session closed for user root  
Sep 02 12:54:02 kali sudo(26369): pam_unix(sudo:session): session opened for user root(uid=0) by kali(uid=1000)  
Sep 02 12:54:03 kali sudo(26369): pam_unix(sudo:session): session closed for user root  
Sep 02 12:54:36 kali sudo(27135): pam_unix(sudo:session): session opened for user root(uid=0) by kali(uid=1000)  
Sep 02 12:54:37 kali sudo(27135): pam_unix(sudo:session): session closed for user root  
Sep 02 12:55:01 kali sudo(27457): pam_unix(sudo:session): session opened for user root(uid=0) by kali(uid=1000)  
Sep 02 12:55:01 kali CRON(27462): pam_unix(cron:session): session opened for user root(uid=0) by root(uid=0)  
Sep 02 12:55:01 kali CRON(27462): pam_unix(cron:session): session closed for user root  
Sep 02 12:55:01 kali sudo(27457): pam_unix(sudo:session): session closed for user root  
Sep 02 12:55:24 kali sudo(28202): pam_unix(sudo:session): session opened for user root(uid=0) by kali(uid=1000)  
Sep 02 12:55:24 kali sudo(28202): pam_unix(sudo:session): session closed for user root  
Sep 02 12:55:33 kali sudo(28298): pam_unix(sudo:session): session opened for user root(uid=0) by kali(uid=1000)  
Sep 02 12:55:33 kali sudo(28298): pam_unix(sudo:session): session closed for user root  
Sep 02 12:56:23 kali sudo(29103): pam_unix(sudo:session): session opened for user root(uid=0) by kali(uid=1000)  
Sep 02 12:56:23 kali sudo(29103): pam_unix(sudo:session): session closed for user root  
Sep 02 12:56:36 kali sudo(29211): pam_unix(sudo:session): session opened for user root(uid=0) by kali(uid=1000)  
Sep 02 12:56:36 kali sudo(29211): pam_unix(sudo:session): session closed for user root  
Sep 02 12:57:11 kali sudo(29919): pam_unix(sudo:session): session opened for user root(uid=0) by kali(uid=1000)  
Sep 02 12:57:12 kali sudo(29919): pam_unix(sudo:session): session closed for user root  
Sep 02 12:57:21 kali sudo(30011): pam_unix(sudo:session): session opened for user root(uid=0) by kali(uid=1000)  
Sep 02 12:57:21 kali sudo(30011): pam_unix(sudo:session): session closed for user root  
Sep 02 12:57:37 kali sudo(30264): pam_unix(sudo:session): session opened for user root(uid=0) by kali(uid=1000)  
Sep 02 12:57:38 kali sudo(30264): pam_unix(sudo:session): session closed for user root  
Sep 02 12:57:58 kali sudo(30454): pam_unix(sudo:session): session opened for user root(uid=0) by kali(uid=1000)  
Sep 02 12:57:58 kali sudo(30454): pam_unix(sudo:session): session closed for user root  
Sep 02 12:58:02 kali sudo(30484): pam_unix(sudo:session): session opened for user root(uid=0) by kali(uid=1000)  
Sep 02 12:58:02 kali sudo(30484): pam_unix(sudo:session): session closed for user root  
Sep 02 13:00:08 kali sudo(32099): pam_unix(sudo:session): session opened for user root(uid=0) by kali(uid=1000)  
  
(kali@kali): ~  
$
```

Figure 12: Live session identification using who, w, and users commands.

```
kali@kali: ~  
Session Actions Edit View Help  
  
(kali@kali): ~  
$ sudo journalctl --no-pager | grep -Ei "sudo"  
Jul 19 11:17:21 kali sudo(27784): kali : TTY=pts/0 ; PWD=/home/kali ; USER=root ; COMMAND=/usr/bin/rm hi  
Jul 19 11:17:21 kali sudo(27784): pam_unix(sudo:session): session opened for user root(uid=0) by kali(uid=1000)  
Jul 19 11:17:21 kali sudo(27784): pam_unix(sudo:session): session closed for user root  
Jul 19 11:27:08 kali sudo(32862): kali : TTY=pts/0 ; PWD=/home/kali ; USER=root ; COMMAND=/usr/bin/su  
Jul 19 11:27:08 kali sudo(32862): pam_unix(sudo:session): session opened for user root(uid=0) by kali(uid=1000)  
Jul 19 11:27:19 kali sudo(32862): pam_unix(sudo:session): session closed for user root  
Jul 19 11:27:26 kali sudo(33061): kali : TTY=pts/0 ; PWD=/home/kali ; USER=root ; COMMAND=/usr/bin/su  
Jul 19 11:27:26 kali sudo(33061): pam_unix(sudo:session): session opened for user root(uid=0) by kali(uid=1000)  
Jul 19 11:30:40 kali sudo(33061): pam_unix(sudo:session): session closed for user root  
Jul 19 11:34:42 kali sudo(36931): kali : TTY=pts/0 ; PWD=/home/kali/Desktop ; USER=root ; COMMAND=/usr/bin/nano hii  
Jul 19 11:34:42 kali sudo(36931): pam_unix(sudo:session): session opened for user root(uid=0) by kali(uid=1000)  
Jul 19 11:34:57 kali sudo(36931): pam_unix(sudo:session): session closed for user root  
Jul 19 11:35:49 kali sudo(37580): kali : TTY=pts/0 ; PWD=/home/kali/Desktop ; USER=root ; COMMAND=/usr/bin/rm hii  
Jul 19 11:35:49 kali sudo(37580): pam_unix(sudo:session): session opened for user root(uid=0) by kali(uid=1000)  
Jul 19 11:35:49 kali sudo(37580): pam_unix(sudo:session): session closed for user root  
Jul 20 05:32:56 kali sudo(20621): kali : TTY=pts/0 ; PWD=/home/kali ; USER=root ; COMMAND=/usr/bin/apt install http  
Jul 20 05:32:56 kali sudo(20621): pam_unix(sudo:session): session opened for user root(uid=0) by kali(uid=1000)  
Jul 20 05:32:57 kali sudo(20621): pam_unix(sudo:session): session closed for user root  
Jul 20 05:33:57 kali sudo(21116): kali : TTY=pts/0 ; PWD=/home/kali ; USER=root ; COMMAND=/usr/bin/apt install hing3  
Jul 20 05:33:57 kali sudo(21116): pam_unix(sudo:session): session opened for user root(uid=0) by kali(uid=1000)  
Jul 20 05:33:57 kali sudo(21116): pam_unix(sudo:session): session closed for user root  
Jul 20 05:35:13 kali sudo(21725): kali : TTY=pts/0 ; PWD=/home/kali ; USER=root ; COMMAND=/usr/bin/su  
Jul 20 05:35:13 kali sudo(21725): pam_unix(sudo:session): session opened for user root(uid=0) by kali(uid=1000)  
Jul 20 05:36:07 kali sudo(22206): root : TTY=pts/2 ; PWD=/home/kali/Desktop ; USER=root ; COMMAND=/usr/bin/apt install hping3  
Jul 20 05:36:07 kali sudo(22206): pam_unix(sudo:session): session opened for user root(uid=0) by kali(uid=0)  
Jul 20 05:36:07 kali sudo(22206): pam_unix(sudo:session): session closed for user root  
Jul 20 05:37:47 kali sudo(23022): root : TTY=pts/2 ; PWD=/home/kali/Desktop ; USER=root ; COMMAND=/usr/sbin/hping3 --flood -S-p 80192.168.200.48  
Jul 20 05:37:47 kali sudo(23022): pam_unix(sudo:session): session opened for user root(uid=0) by kali(uid=0)  
Jul 20 05:37:47 kali sudo(23022): pam_unix(sudo:session): session closed for user root  
Jul 20 05:37:53 kali sudo(23068): root : TTY=pts/2 ; PWD=/home/kali/Desktop ; USER=root ; COMMAND=/usr/sbin/hping3 --flood -S-p 80192.168.200.148
```

Figure 13: Supplementary live session verification confirming terminal line seats.

13. AUTHENTICATION LOG ANALYSIS

Systemd journal inspection, socket listening bindings, and privilege auditing logs:

```
kali@kali: ~  
Session Actions Edit View Help  
[kali@kali]~$ sudo ss -tulpn | grep ':22'  
tcp    LISTEN 0      128          0.0.0.0:*    0.0.0.0:*    users:({sshd,pid=650,fd=6})  
tcp    LISTEN 0      128          [::]:22     [::]:*      users:({sshd,pid=650,fd=7})  
[kali@kali]~$
```

Figure 14: Systemd verification showing active running state for ssh.service.

```
kali@kali: ~  
Session Actions Edit View Help  
[kali@kali]~$ cat ~/Project2_Evidence/system_information.txt  
OS  
PRETTY_NAME="Kali GNU/Linux Rolling"  
NAME="Kali GNU/Linux"  
VERSION_ID="2026.2"  
VERSION="2026.2"  
VERSION_CODENAME=kali-rolling  
ID=kali  
ID_LIKE=debian  
HOME_URL="https://www.kali.org/"  
SUPPORT_URL="https://forums.kali.org/"  
BUG_REPORT_URL="https://bugs.kali.org/"  
ANSI_COLOR="1;31"  
  
===== KERNEL =====  
Linux kali 6.19.14+kali-amd64 #1 SMP PREEMPT_DYNAMIC Kali 6.19.14-1+kali1 (2026-05-05) x86_64 GNU/Linux  
  
===== HOST =====  
Static hostname: kali  
Icon name: computer-vn  
Chassis: vm  
Machine ID: 246bd9421a9a4c0d9da68d5f5abcfabcd  
Boot ID: 1d0a36a2a77b49bd935a83cee05e1f82  
Virtualization: oracle  
Operating System: kali GNU/Linux Rolling  
Kernel: Linux 6.19.14+kali-amd64  
Architecture: x86-64  
Hardware Vendor: innotek GmbH  
Hardware Model: VirtualBox  
Hardware Version: 1.2  
Firmware Version: VirtualBox  
Firmware Date: Fri 2006-12-01  
Firmware Age: 19y 9month 2d  
  
===== DATE =====  
Wed Sep 2 02:12:48 PM EDT 2026
```

Figure 15: Network socket auditing confirming sshd listening parameters.

```
kali@kali: ~  
Session Actions Edit View Help  
[kali@kali]~  
$ sudo journalctl -u ssh --no-pager -n 100  
Sep 01 12:04:48 kali systemd[1]: Starting ssh.service - OpenBSD Secure Shell server...  
Sep 01 12:04:48 kali sshd[33045]: Server listening on 0.0.0.0 port 22.  
Sep 01 12:04:48 kali sshd[33045]: Server listening on :: port 22.  
Sep 01 12:04:48 kali systemd[1]: Started ssh.service - OpenBSD Secure Shell server.  
-- Boot a3345b5a1cb04d01af38d62f654e2e9e --  
Sep 01 12:39:55 kali systemd[1]: Starting ssh.service - OpenBSD Secure Shell server...  
Sep 01 12:39:55 kali sshd[691]: Server listening on 0.0.0.0 port 22.  
Sep 01 12:39:55 kali sshd[691]: Server listening on :: port 22.  
Sep 01 12:39:55 kali systemd[1]: Started ssh.service - OpenBSD Secure Shell server.  
-- Boot aae3dfacf5574f0491bb0f9f9495c61ef --  
Sep 01 12:49:04 kali systemd[1]: Starting ssh.service - OpenBSD Secure Shell server...  
Sep 01 12:49:04 kali sshd[648]: Server listening on 0.0.0.0 port 22.  
Sep 01 12:49:04 kali sshd[648]: Server listening on :: port 22.  
Sep 01 12:49:04 kali systemd[1]: Started ssh.service - OpenBSD Secure Shell server.  
-- Boot 5a5f0d69522842058a841e91c0498dd1 --  
Sep 01 12:51:14 kali systemd[1]: Starting ssh.service - OpenBSD Secure Shell server...  
Sep 01 12:51:14 kali sshd[648]: Server listening on 0.0.0.0 port 22.  
Sep 01 12:51:14 kali sshd[648]: Server listening on :: port 22.  
Sep 01 12:51:14 kali systemd[1]: Started ssh.service - OpenBSD Secure Shell server.  
Sep 01 13:25:45 kali sshd-session[20738]: Accepted password for kali from ::1 port 40612 ssh2  
Sep 01 13:25:46 kali sshd-session[20738]: pam_unix(sshd:session): session opened for user kali(uid=1000) by kali(uid=0)  
Sep 01 13:42:23 kali sshd-session[30851]: pam_unix(sshd:auth): authentication failure; logname= uid=0 euid=0 tty=ssh ruser= rhost=::1 user=kali  
Sep 01 13:42:25 kali sshd-session[30851]: Failed password for kali from ::1 port 60380 ssh2  
Sep 01 13:42:37 kali unix_chkpwd[31014]: password check failed for user (kali)  
Sep 01 13:42:39 kali sshd-session[30851]: Failed password for kali from ::1 port 60380 ssh2  
Sep 01 13:42:43 kali unix_chkpwd[31047]: password check failed for user (kali)  
Sep 01 13:42:45 kali sshd-session[30851]: Failed password for kali from ::1 port 60380 ssh2  
Sep 01 13:44:20 kali sshd[648]: Timeout before authentication for connection from ::1 to ::1, pid = 30851  
Sep 01 13:44:45 kali sshd-session[31615]: Accepted password for kali from 127.0.0.1 port 44070 ssh2  
Sep 01 13:44:45 kali sshd-session[31615]: pam_unix(sshd:session): session opened for user kali(uid=1000) by kali(uid=0)
```

Figure 16: Systemd journal output tracking SSH service connection attempts.

```
kali@kali: ~  
Session Actions Edit View Help  
[kali@kali]~  
$ sudo journalctl --no-pager | grep -E "authentication failure|failed password|invalid user|accepted password|accepted publickey" > ~/Project2_Evidence/Authentication_evidence.txt  
[sudo] password for kali:  
[kali@kali]~  
$ cat ~/Project2_Evidence/Authentication_evidence.txt  
Sep 01 13:25:45 kali sshd-session[20738]: Accepted password for kali from ::1 port 40612 ssh2  
Sep 01 13:42:23 kali sshd-session[30851]: pam_unix(sshd:auth): authentication failure; logname= uid=0 euid=0 tty=ssh ruser= rhost=::1 user=kali  
Sep 01 13:42:25 kali sshd-session[30851]: Failed password for kali from ::1 port 60380 ssh2  
Sep 01 13:42:39 kali sshd-session[30851]: Failed password for kali from ::1 port 60380 ssh2  
Sep 01 13:42:45 kali sshd-session[30851]: Failed password for kali from ::1 port 60380 ssh2  
Sep 01 13:44:45 kali sshd-session[31615]: Accepted password for kali from 127.0.0.1 port 44070 ssh2
```

Figure 17: Pattern matching search for authentication failures and successful logins.

```
kali@kali: ~  
Session Actions Edit View Help  
--(kali@kali)-[~]  
└─$ sudo journalctl -u ssh --no-pager > ~/Project2_Evidence/ssh_evidence.txt  
--(kali@kali)-[~]  
└─$ head -100 ~/Project2_Evidence/ssh_evidence.txt  
Sep 01 12:04:48 kali systemd[1]: Starting ssh.service - OpenBSD Secure Shell server...  
Sep 01 12:04:48 kali sshd[33045]: Server listening on 0.0.0.0 port 22.  
Sep 01 12:04:48 kali sshd[33045]: Server listening on :: port 22.  
Sep 01 12:04:48 kali systemd[1]: Started ssh.service - OpenBSD Secure Shell server.  
-- Boot a3345b5a1cb04d01af38d62f654c2e9e --  
Sep 01 12:39:55 kali systemd[1]: Starting ssh.service - OpenBSD Secure Shell server...  
Sep 01 12:39:55 kali sshd[691]: Server listening on 0.0.0.0 port 22.  
Sep 01 12:39:55 kali sshd[691]: Server listening on :: port 22.  
Sep 01 12:39:55 kali systemd[1]: Started ssh.service - OpenBSD Secure Shell server.  
-- Boot aae3dfacf5574f0691bd9fe9405c61ef --  
Sep 01 12:49:04 kali systemd[1]: Starting ssh.service - OpenBSD Secure Shell server...  
Sep 01 12:49:04 kali sshd[648]: Server listening on 0.0.0.0 port 22.  
Sep 01 12:49:04 kali sshd[648]: Server listening on :: port 22.  
Sep 01 12:49:04 kali systemd[1]: Started ssh.service - OpenBSD Secure Shell server.  
-- Boot 5a5f0d69522842058ae841e91c0498dd1 --  
Sep 01 12:51:14 kali systemd[1]: Starting ssh.service - OpenBSD Secure Shell server...  
Sep 01 12:51:14 kali sshd[648]: Server listening on 0.0.0.0 port 22.  
Sep 01 12:51:14 kali sshd[648]: Server listening on :: port 22.  
Sep 01 12:51:14 kali systemd[1]: Started ssh.service - OpenBSD Secure Shell server.  
Sep 01 13:25:45 kali sshd-session[20738]: Accepted password for kali from ::1 port 40612 ssh2  
Sep 01 13:25:46 kali sshd-session[20738]: pam_unix(sshd:session): session opened for user kali(uid=1000) by kali(uid=0)  
Sep 01 13:42:23 kali sshd-session[30851]: pam_unix(sshd:auth): authentication failure; logname= uid=0 euid=0 tty=ssh ruser= rhost=::1 user=kali  
Sep 01 13:42:25 kali sshd-session[30851]: Failed password for kali from ::1 port 60380 ssh2  
Sep 01 13:42:37 kali unix_chkpwd[31014]: password check failed for user (kali)  
Sep 01 13:42:39 kali sshd-session[30851]: Failed password for kali from ::1 port 60380 ssh2  
Sep 01 13:42:43 kali unix_chkpwd[31047]: password check failed for user (kali)  
Sep 01 13:42:45 kali sshd-session[30851]: Failed password for kali from ::1 port 60380 ssh2  
Sep 01 13:44:20 kali sshd[648]: Timeout before authentication for connection from ::1 to ::1, pid = 30851
```

Figure 18: Head inspection of saved SSH evidence text logs.

```
kali@kali: ~  
Session Actions Edit View Help  
--(kali@kali)-[~]  
└─$ sudo journalctl --list-boots  
IDX BOOT ID FIRST ENTRY LAST ENTRY  
-47 a3c458792f53431794b865cb6e77221c Tue 2026-07-07 06:17:09 EDT Thu 2026-07-16 06:01:36 EDT  
-46 c2a20a1ebba248539a790e6ea39f3d82 Fri 2026-07-17 02:04:35 EDT Sun 2026-07-19 11:55:19 EDT  
-45 f854191a83b54afb8faa575254f6dfa8 Mon 2026-07-20 04:54:32 EDT Mon 2026-07-20 06:03:10 EDT  
-44 6acd9c794d214e39b425eed4c48a1d0d Mon 2026-07-20 06:04:17 EDT Mon 2026-07-20 06:06:16 EDT  
-43 5923594c899549e0ac065a0f6ef906b0 Mon 2026-07-20 06:08:29 EDT Mon 2026-07-20 06:19:34 EDT  
-42 c7cb6adb0ebf465d94b5447f2c3f2105 Mon 2026-07-20 06:21:05 EDT Mon 2026-07-20 06:52:29 EDT  
-41 21546d2cdf49a39691e149ada14bf2 Mon 2026-07-20 06:53:58 EDT Mon 2026-07-20 06:56:45 EDT  
-40 9f10eade9f94c48deb752b9445fa6710e Mon 2026-07-20 06:58:01 EDT Mon 2026-07-20 07:00:35 EDT  
-39 4cc42680ce433aaac5525ba64d90a Mon 2026-07-20 07:02:50 EDT Mon 2026-07-20 07:28:53 EDT  
-38 b38bd0e29b084d79bfcdab9701a18fb9 Mon 2026-07-20 07:31:10 EDT Mon 2026-07-20 07:50:30 EDT  
-37 11fcaeb0a56447d9ec76a02e02040d1 Mon 2026-07-20 07:56:15 EDT Mon 2026-07-20 11:22:18 EDT  
-36 fde9b13ba18b4fd382524a8db91bcc9 Mon 2026-07-20 11:25:37 EDT Tue 2026-07-21 07:17:02 EDT  
-35 80dfcae6b19b445b06f24c405893b13f Tue 2026-07-21 07:26:48 EDT Tue 2026-07-21 08:01:34 EDT  
-34 af7ec56d5eca4dbf9046a63bd2150519 Wed 2026-07-22 04:49:38 EDT Wed 2026-07-22 05:25:01 EDT  
-33 17ca938633a9486fb96276d8d52c1200 Wed 2026-07-22 10:53:09 EDT Wed 2026-07-22 10:53:40 EDT  
-32 4ad178c7a2a24c26a0c04c5fa0e08e8 Thu 2026-07-23 01:18:00 EDT Thu 2026-07-23 02:45:01 EDT  
-31 3372ad1d1f2242979c09c49d95d3aad Thu 2026-07-23 04:46:27 EDT Thu 2026-07-23 04:48:24 EDT  
-30 2da082967ad4e7b85fb2a1f62c58c5f Thu 2026-07-23 12:04:48 EDT Thu 2026-07-23 12:55:01 EDT  
-29 b13efde88bb4096bfc56d10bdacfb1 Mon 2026-07-27 04:37:09 EDT Mon 2026-07-27 04:39:02 EDT  
-28 85c4cefebed64317ab5b059edc4a1fef Mon 2026-07-27 11:42:37 EDT Wed 2026-07-29 13:17:01 EDT  
-27 6328fe389e954eazb77e27f7e88e9cee Thu 2026-07-30 05:41:49 EDT Thu 2026-07-30 05:59:53 EDT  
-26 2445082206f4da9b9c3323fe7e00075e Thu 2026-07-30 06:03:50 EDT Thu 2026-07-30 06:04:28 EDT  
-25 a54701d5d82d4d9788d86d5024c7b6e0 Thu 2026-07-30 06:06:10 EDT Thu 2026-07-30 13:39:01 EDT  
-24 f9e55b9a558945f285930ef72dc676e5 Fri 2026-07-31 01:14:39 EDT Fri 2026-07-31 01:39:02 EDT  
-23 d18552db140742d8abeee90d463c3cf Fri 2026-07-31 02:05:25 EDT Fri 2026-07-31 03:10:01 EDT  
-22 dc751bf0071a40afbdf3e9b1f874373f Sun 2026-08-02 02:49:31 EDT Sun 2026-08-02 03:05:01 EDT  
-21 a3bc4f214d69450e840016c700cd57f Sun 2026-08-02 08:13:18 EDT Thu 2026-08-06 02:25:01 EDT  
-20 75057f603594caf8bf75a50214cae0 Thu 2026-08-06 07:37:32 EDT Thu 2026-08-06 08:05:01 EDT  
-19 f5fe433fe5214eb893e69626d2ad88a2 Mon 2026-08-10 05:31:24 EDT Tue 2026-08-11 11:14:02 EDT
```

Figure 19: Boot index tracking via journalctl --list-boots.


```
kali@kali: ~  
Session Actions Edit View Help  
[kali@kali]~  
$ sudo journalctl --no-pager | grep -Ei "boot|shutdown|reboot"  
Jul 07 06:17:09 kali kernel: Command line: BOOT_IMAGE=/boot/vmlinuz-6.19.14+kali-amd64 root=UUID=d8f71766-9813-406b-90f2-fd6cd673b79e ro quiet splash  
Jul 07 06:17:09 kali kernel: Booting paravirtualized kernel on KVM  
Jul 07 06:17:09 kali kernel: Kernel command line: BOOT_IMAGE=/boot/vmlinuz-6.19.14+kali-amd64 root=UUID=d8f71766-9813-406b-90f2-fd6cd673b79e ro quiet splash  
Jul 07 06:17:09 kali kernel: smpboot: CPU0: Intel(R) Core(TM) i5-7300U CPU @ 2.60GHz (family: 0x6, model: 0x8e, stepping: 0x9)  
Jul 07 06:17:09 kali kernel: smpboot: x86: Running SMP configuration:  
Jul 07 06:17:09 kali kernel: smpboot: Total of 2 processors activated (10847.98 BogoMIPS)  
Jul 07 06:17:09 kali kernel: pci 0000:00:02.0: vgaarb: setting as boot VGA device  
Jul 07 06:17:09 kali systemd[1]: Starting ifupdown-pre.service - Helper to synchronize boot up for ifupdown ...  
Jul 07 06:17:10 kali systemd[1]: Finished ifupdown-pre.service - Helper to synchronize boot up for ifupdown.  
Jul 07 06:17:11 kali systemd[1]: systemd-firstboot.service - Initial Setup skipped, unmet condition check ConditionFirstBoot=yes  
Jul 07 06:17:11 kali systemd[1]: First boot complete, target - First boot Complete skipped, unmet condition check ConditionFirstBoot=yes  
Jul 07 06:17:12 kali systemd[1]: Starting plymouth-start.service - Show Plymouth Boot Screen ...  
Jul 07 06:17:12 kali systemd[1]: Started plymouth-start.service - Show Plymouth Boot Screen.  
Jul 07 06:17:30 kali cron[738]: (CRON) INFO (Running @reboot jobs)  
Jul 07 06:17:31 kali NetworkManager[769]: <info> [1783419451.4419] NetworkManager (version 1.56.1) is starting ... (boot:a3c45879-2f53-4317-94b8-65cb6e77221c)  
Jul 07 06:17:32 kali systemd[1]: Starting plymouth-quit-wait.service - Hold until boot process finishes up ...  
Jul 07 06:17:32 kali systemd[1]: Finished plymouth-quit-wait.service - Hold until boot process finishes up.  
Jul 07 06:18:00 kali systemd[902]: Reached target shutdown.target - Shutdown.  
-- Boot c2a20a1ebba248539a790e6ea39f3d82 --  
-- Boot a3c458792f53431794b865cb6e77221c --  
-- Boot c2a20a1ebba248539a790e6ea39f3d82 --  
Jul 17 02:04:35 kali kernel: Command line: BOOT_IMAGE=/boot/vmlinuz-6.19.14+kali-amd64 root=UUID=d8f71766-9813-406b-90f2-fd6cd673b79e ro quiet splash  
-- Boot a3c458792f53431794b865cb6e77221c --  
-- Boot c2a20a1ebba248539a790e6ea39f3d82 --  
-- Boot a3c458792f53431794b865cb6e77221c --  
-- Boot c2a20a1ebba248539a790e6ea39f3d82 --  
-- Boot a3c458792f53431794b865cb6e77221c --  
-- Boot c2a20a1ebba248539a790e6ea39f3d82 --  
-- Boot a3c458792f53431794b865cb6e77221c --  
-- Boot c2a20a1ebba248539a790e6ea39f3d82 --  
-- Boot a3c458792f53431794b865cb6e77221c --  
-- Boot c2a20a1ebba248539a790e6ea39f3d82 --
```

Figure 20: System startup and shutdown sequence logs.

```
kali@kali: ~  
Session Actions Edit View Help  
[kali@kali]~  
$ sudo journalctl --no-pager | grep -Ei "sudo" > ~/Project2_Evidence/sudo_evidence.txt  
[kali@kali]~  
$ cat ~/Project2_Evidence/sudo_evidence.txt  
Jul 19 11:17:21 kali sudo[27784]: kali : TTY=pts/0 ; PWD=/home/kali ; USER=root ; COMMAND=/usr/bin/rm hi  
Jul 19 11:17:21 kali sudo[27784]: pam_unix(sudo:session): session opened for user root(uid=0) by kali(uid=1000)  
Jul 19 11:17:21 kali sudo[27784]: pam_unix(sudo:session): session closed for user root  
Jul 19 11:27:08 kali sudo[32862]: kali : TTY=pts/0 ; PWD=/home/kali ; USER=root ; COMMAND=/usr/bin/su  
Jul 19 11:27:08 kali sudo[32862]: pam_unix(sudo:session): session opened for user root(uid=0) by kali(uid=1000)  
Jul 19 11:27:19 kali sudo[32862]: pam_unix(sudo:session): session closed for user root  
Jul 19 11:27:26 kali sudo[33061]: kali : TTY=pts/0 ; PWD=/home/kali ; USER=root ; COMMAND=/usr/bin/su  
Jul 19 11:27:26 kali sudo[33061]: pam_unix(sudo:session): session opened for user root(uid=0) by kali(uid=1000)  
Jul 19 11:30:40 kali sudo[33061]: pam_unix(sudo:session): session closed for user root  
Jul 19 11:34:42 kali sudo[36931]: kali : TTY=pts/0 ; PWD=/home/kali/Desktop ; USER=root ; COMMAND=/usr/bin/nano hii  
Jul 19 11:34:42 kali sudo[36931]: pam_unix(sudo:session): session opened for user root(uid=0) by kali(uid=1000)  
Jul 19 11:34:57 kali sudo[36931]: pam_unix(sudo:session): session closed for user root  
Jul 19 11:35:49 kali sudo[37580]: kali : TTY=pts/0 ; PWD=/home/kali/Desktop ; USER=root ; COMMAND=/usr/bin/rm hii  
Jul 19 11:35:49 kali sudo[37580]: pam_unix(sudo:session): session opened for user root(uid=0) by kali(uid=1000)  
Jul 19 11:35:49 kali sudo[37580]: pam_unix(sudo:session): session closed for user root  
Jul 20 05:32:56 kali sudo[20621]: kali : TTY=pts/0 ; PWD=/home/kali ; USER=root ; COMMAND=/usr/bin/apt install htop  
Jul 20 05:32:56 kali sudo[20621]: pam_unix(sudo:session): session opened for user root(uid=0) by kali(uid=1000)  
Jul 20 05:32:57 kali sudo[20621]: pam_unix(sudo:session): session closed for user root  
Jul 20 05:33:57 kali sudo[21116]: kali : TTY=pts/0 ; PWD=/home/kali ; USER=root ; COMMAND=/usr/bin/apt install hings3  
Jul 20 05:33:57 kali sudo[21116]: pam_unix(sudo:session): session opened for user root(uid=0) by kali(uid=1000)  
Jul 20 05:33:57 kali sudo[21116]: pam_unix(sudo:session): session closed for user root  
Jul 20 05:35:13 kali sudo[21725]: kali : TTY=pts/0 ; PWD=/home/kali ; USER=root ; COMMAND=/usr/bin/su  
Jul 20 05:35:13 kali sudo[21725]: pam_unix(sudo:session): session opened for user root(uid=0) by kali(uid=1000)  
Jul 20 05:36:07 kali sudo[22206]: root : TTY=pts/2 ; PWD=/home/kali/Desktop ; USER=root ; COMMAND=/usr/bin/apt install hping3  
Jul 20 05:36:07 kali sudo[22206]: pam_unix(sudo:session): session opened for user root(uid=0) by kali(uid=0)  
Jul 20 05:36:07 kali sudo[22206]: pam_unix(sudo:session): session closed for user root  
Jul 20 05:37:47 kali sudo[23022]: root : TTY=pts/2 ; PWD=/home/kali/Desktop ; USER=root ; COMMAND=/usr/sbin/hping3 --flood -S-p 80192.168.200.48
```

Figure 21: Filtered sudo and su execution logs.

```
kali@kali: ~  
Session Actions Edit View Help  
kali tty7 :0 Mon Jul 20 11:26 - still logged in  
lightdm tty7 :0 Mon Jul 20 11:25 - 11:26 (00:00)  
kali tty7 :0 Mon Jul 20 07:56 - still logged in  
lightdm tty7 :0 Mon Jul 20 07:56 - 07:56 (00:00)  
kali tty7 :0 Mon Jul 20 07:32 - still logged in  
lightdm tty7 :0 Mon Jul 20 07:31 - 07:32 (00:00)  
kali tty7 :0 Mon Jul 20 07:03 - still logged in  
lightdm tty7 :0 Mon Jul 20 07:02 - 07:03 (00:00)  
kali tty7 :0 Mon Jul 20 06:58 - still logged in  
lightdm tty7 :0 Mon Jul 20 06:58 - 06:58 (00:00)  
kali tty7 :0 Mon Jul 20 06:54 - still logged in  
lightdm tty7 :0 Mon Jul 20 06:54 - 06:54 (00:00)  
kali tty7 :0 Mon Jul 20 06:21 - still logged in  
lightdm tty7 :0 Mon Jul 20 06:21 - 06:21 (00:00)  
kali tty7 :0 Mon Jul 20 06:08 - still logged in  
lightdm tty7 :0 Mon Jul 20 06:08 - 06:08 (00:00)  
kali tty7 :0 Mon Jul 20 06:04 - still logged in  
lightdm tty7 :0 Mon Jul 20 06:04 - 06:04 (00:00)  
root pts/2 :0 Mon Jul 20 05:35 - still logged in  
kali tty7 :0 Mon Jul 20 04:54 - still logged in  
lightdm tty7 :0 Mon Jul 20 04:54 - 04:54 (00:00)  
root pts/1 :0 Sun Jul 19 11:27 - 11:30 (00:03)  
root pts/1 :0 Sun Jul 19 11:27 - 11:27 (00:00)  
kali tty7 :0 Fri Jul 17 02:04 - still logged in  
lightdm tty7 :0 Fri Jul 17 02:04 - 02:06 (00:00)  
kali tty7 :0 Tue Jul 7 06:17 - still logged in  
lightdm tty7 :0 Tue Jul 7 06:17 - 06:17 (00:00)  
postgres :0 Tue Jun 16 10:29 - 10:29 (00:00)  
  
wtmpdb begins Tue Jun 16 10:29:31 2026  
  
kali@kali:~$
```

Figure 22: Additional chronological audit entries showing administrative commands.

```
kali@kali: ~  
Session Actions Edit View Help  
  
kali@kali:~$ sudo journalctl --no-pager | grep -E "session opened|session closed"  
Jul 07 06:17:36 kali lightdm[883]: pam_unix(lightdm-greeter:session): session opened for user lightdm(uid=106) by (uid=0)  
Jul 07 06:17:36 kali (systemd)[902]: pam_unix(systemd-user:session): session opened for user lightdm(uid=106) by lightdm(uid=0)  
Jul 07 06:17:37 kali lightdm[883]: pam_unix(lightdm-greeter:session): session opened for user lightdm(uid=106) by (uid=0)  
Jul 07 06:17:48 kali lightdm[883]: pam_unix(lightdm-greeter:session): session closed for user lightdm  
Jul 07 06:17:48 kali lightdm[883]: pam_unix(lightdm-greeter:session): session closed for user lightdm  
Jul 07 06:17:48 kali lightdm[1062]: pam_unix(lightdm:session): session opened for user kali(uid=1000) by kali(uid=0)  
Jul 07 06:17:48 kali (systemd)[1006]: pam_unix(systemd-user:session): session opened for user kali(uid=1000) by kali(uid=0)  
Jul 07 02:04:44 kali lightdm[717]: pam_unix(lightdm-greeter:session): session opened for user lightdm(uid=106) by (uid=0)  
Jul 07 02:04:44 kali (systemd)[725]: pam_unix(systemd-user:session): session opened for user lightdm(uid=106) by lightdm(uid=0)  
Jul 07 02:04:45 kali lightdm[717]: pam_unix(lightdm-greeter:session): session opened for user lightdm(uid=106) by (uid=0)  
Jul 07 02:04:53 kali lightdm[717]: pam_unix(lightdm-greeter:session): session closed for user lightdm  
Jul 07 02:04:53 kali lightdm[883]: pam_unix(lightdm:session): session opened for user kali(uid=1000) by kali(uid=0)  
Jul 07 02:04:53 kali (systemd)[907]: pam_unix(systemd-user:session): session opened for user kali(uid=1000) by kali(uid=0)  
Jul 07 02:05:01 kali CRON[1547]: pam_unix(cron:session): session opened for user root(uid=0) by root(uid=0)  
Jul 07 02:05:01 kali CRON[1547]: pam_unix(cron:session): session closed for user root  
Jul 07 02:05:04 kali (sd-pam)[727]: pam_unix(systemd-user:session): session closed for user lightdm  
Jul 07 02:09:01 kali CRON[4054]: pam_unix(cron:session): session opened for user root(uid=0) by root(uid=0)  
Jul 07 02:09:01 kali CRON[4054]: pam_unix(cron:session): session closed for user root  
Jul 07 02:15:01 kali CRON[7383]: pam_unix(cron:session): session opened for user root(uid=0) by root(uid=0)  
Jul 07 02:15:01 kali CRON[7383]: pam_unix(cron:session): session closed for user root  
Jul 07 02:17:01 kali CRON[8336]: pam_unix(cron:session): session opened for user root(uid=0) by root(uid=0)  
Jul 07 02:17:01 kali CRON[8336]: pam_unix(cron:session): session closed for user root  
Jul 07 07:55:01 kali CRON[12703]: pam_unix(cron:session): session opened for user root(uid=0) by root(uid=0)  
Jul 07 07:55:01 kali CRON[12703]: pam_unix(cron:session): session closed for user root  
Jul 07 08:05:01 kali CRON[17829]: pam_unix(cron:session): session opened for user root(uid=0) by root(uid=0)  
Jul 07 08:05:01 kali CRON[17829]: pam_unix(cron:session): session closed for user root  
Jul 07 08:09:01 kali CRON[19821]: pam_unix(cron:session): session opened for user root(uid=0) by root(uid=0)  
Jul 07 08:09:01 kali CRON[19821]: pam_unix(cron:session): session closed for user root  
Jul 07 08:15:02 kali CRON[22908]: pam_unix(cron:session): session opened for user root(uid=0) by root(uid=0)
```

Figure 23: Regular expression filtered log dump tracking session open and close states.

```
kali@kali: ~  
$ sudo wtmpdo last  
kali tty7 :0 Wed Sep 2 12:21 - still logged in  
lightdm tty7 :0 Wed Sep 2 12:21 - 12:21 (00:00)  
kali tty7 :0 Wed Sep 2 11:31 - still logged in  
lightdm tty7 :0 Wed Sep 2 11:31 - 11:31 (00:00)  
kali tty7 :0 Wed Sep 2 03:00 - still logged in  
lightdm tty7 :0 Wed Sep 2 02:50 - 03:00 (00:00)  
kali tty7 :0 Tue Sep 1 23:24 - still logged in  
lightdm tty7 :0 Tue Sep 1 23:23 - 23:24 (00:01)  
kali pts/4 127.0.0.1 Tue Sep 1 13:44 - still logged in  
kali pts/3 ::1 Tue Sep 1 13:25 - still logged in  
kali tty7 :0 Tue Sep 1 12:51 - still logged in  
lightdm tty7 :0 Tue Sep 1 12:51 - 12:51 (00:00)  
kali tty7 :0 Tue Sep 1 12:49 - still logged in  
lightdm tty7 :0 Tue Sep 1 12:49 - 12:49 (00:00)  
kali tty7 :0 Tue Sep 1 12:40 - still logged in  
lightdm tty7 :0 Tue Sep 1 12:40 - 12:40 (00:00)  
kali tty7 :0 Tue Sep 1 06:16 - still logged in  
lightdm tty7 :0 Tue Sep 1 06:15 - 06:16 (00:00)  
kali tty7 :0 Thu Aug 27 11:26 - still logged in  
lightdm tty7 :0 Thu Aug 27 11:26 - 11:26 (00:00)  
kali tty7 :0 Thu Aug 27 11:21 - still logged in  
lightdm tty7 :0 Thu Aug 27 11:21 - 11:21 (00:00)  
kali tty7 :0 Tue Aug 18 11:41 - still logged in  
lightdm tty7 :0 Tue Aug 18 11:21 - 11:41 (00:20)  
kali tty7 :0 Tue Aug 18 00:08 - still logged in  
lightdm tty7 :0 Tue Aug 18 00:07 - 00:08 (00:00)  
kali tty7 :0 Mon Aug 17 11:33 - still logged in  
lightdm tty7 :0 Mon Aug 17 11:33 - 11:33 (00:00)  
kali tty7 :0 Mon Aug 17 07:24 - still logged in  
lightdm tty7 :0 Mon Aug 17 07:21 - 07:24 (00:02)
```

Figure 24: Systemd PAM session life-cycle logs.

14. AUTHENTICATION EVENT INTERPRETATION

Analysis of loopback address differences (::1 vs 127.0.0.1) and dual-stack client fallback behaviors.

15. POTENTIALLY SUSPICIOUS ACTIVITY

Evaluation of local loopback failed password attempts and risk assessment (Low risk / False positive for external intrusion).

16. INVESTIGATION FINDINGS

Comprehensive summary of environment settings, current session state, log structures, and forensic results.

17. CHALLENGES FACED

Adapting traditional log analysis workflows to SQLite databases, timestamp reconciliation, and high-volume filtering.

18. LEARNING OUTCOMES

Mastery over Linux authentication sub-systems, sqlite3 inspection, journalctl filtering, and threat pattern recognition.

19. FUTURE SCOPE

Automated log analysis pipelines, SIEM integration, real-time anomaly alerts, and Y2038 compliance testing.

20. CONCLUSION

The project successfully achieved all technical objectives, confirming that observed SSH failures were benign local tests.

21. REFERENCES

1. Kali Linux Documentation: System Logging and Security Auditing Standards.
2. Man Pages: wtmpdb(8), last(1), sqlite3(1), journalctl(1), sshd(8).
3. Systemd Documentation.
4. SQLite Documentation.